Government LLMiner Project Update and Demo

04.12.2025

Project Context and Impact Pathway on AI and DPI Trajectory : *LLM-powered Intelligence for Governance Decision-making to support evidence-based decision and equitable service delivery*

Solving Fragmented Governance Insight especially in unstructured data

Governments struggle with siloed systems, limited data literacy, and scattered unstructured information that hide critical facts. This platform unifies documents, reports, interviews, community feedback, and operational logs into a single reasoning engine that extracts meaning, highlights risks, exposes inefficiencies, and supports faster accountable decision making.

Theory of impact is a better resource management through unified evidence

When insights are triangulated across ministries and time, government decision makers can easily identify policy bias, resource leakage, underserved communities, and opportunities for improvement. Benchmarking, early warning analytics, trend detection, and impact measurement all strengthen transparency, citizen trust, and data-driven service delivery that improves lives where infrastructure is weakest.

Deployment approach that sustainability focussed with low maintenance cost

An offline-first deployment eliminates cloud dependence, reduces cost, and protects sensitive public data from external exposure. Local GPU and servers enable continuous model retraining, uninterrupted access in low-connectivity environments, and long-term sustainability through local support, simplified maintenance, and reduced recurring spend on compute and bandwidth.

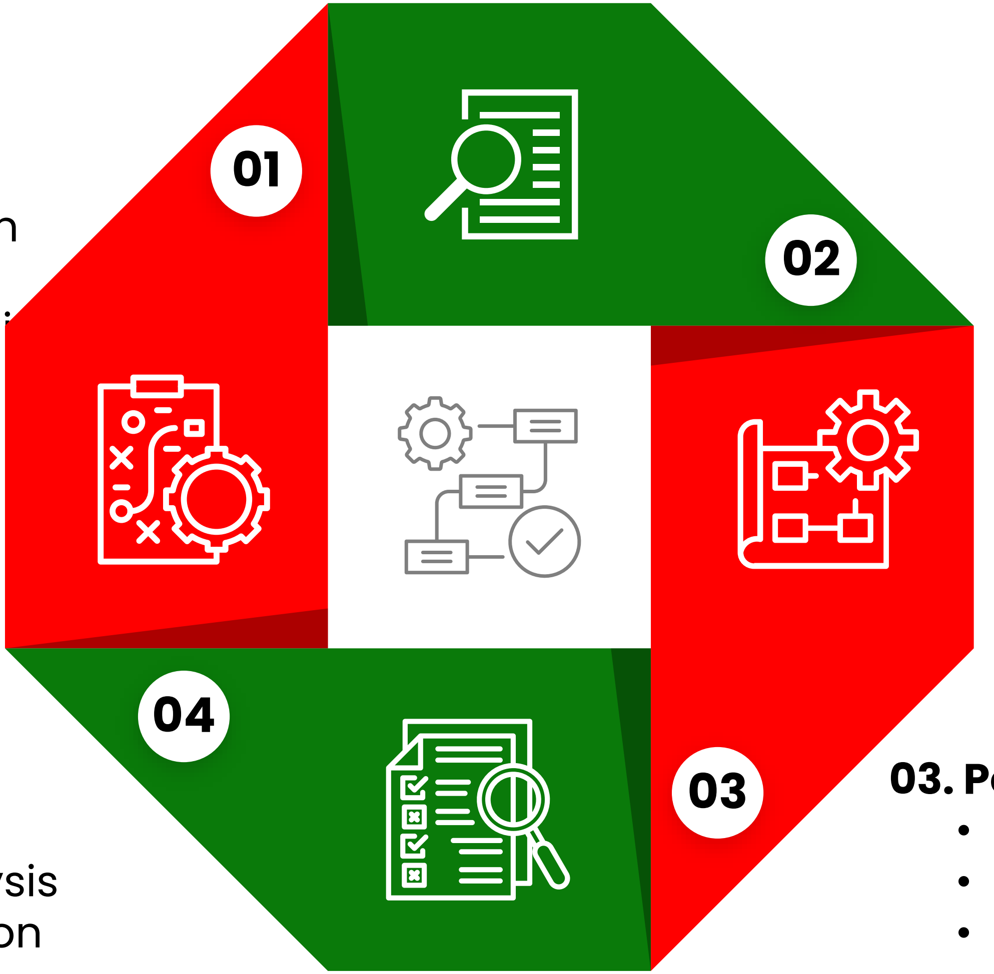
What makes it better than any LLM - Governance-Tuned Intelligence System

Unlike generic LLMs, it embeds predefined governance prompt templates, domain ontologies, and Retrieval Augmented Generation for contextual accuracy from verified local data. It reasons using government language, policies, KPI structures, and socio-economic frameworks, delivering precise actionable insights aligned with public sector accountability and equity priorities.

Extensive engagement with states to identify Government LLM Use Cases Mapped to Government Policy Workflow based on learnings from Singapore

01. Policy Formulation

- Bias Detection and Inclusion Gap Analysis in document
- Thematic and Trend Analysi
- Inter-agency Consistency Check



02. Policy Review

- Regulatory Gap Identification
- Document Comparison and Evolution Tracking
- Narrative Framing and Sentiment Analysis

04. Policy Evaluation

- Policy Impact Traceability
- Thematic and Trend Analysis
- Bias Detection and Inclusion Gap Analysis

03. Policy Implementation

- Semantic Search and Retrieval
- Compliance & Legal Risk Analysis
- Data Extraction for Structured Policy
- Dashboards

AIKano End-to-End Solution Design Framework

Model & Reasoning Stack

- Open-source model (Qwen) and Kano State -tuned Small Language Model reason over internal knowledge using RAG, ensuring responses are contextual, accurate, and aligned with institutional realities with ethno-religious guardrails.

On-Premise/Offline GPU Infrastructure

- BIZON X5500 enables offline training and inference for 0.5–1.3B-parameter models, processing up to 1B tokens daily, securing full data control and high-performance operations.

Secure Graduated Access

- Role-based permissions, strict LAN isolation and usage audits that ensure only authorised staff access sensitive insights while maintaining compliance, traceability, and information integrity.

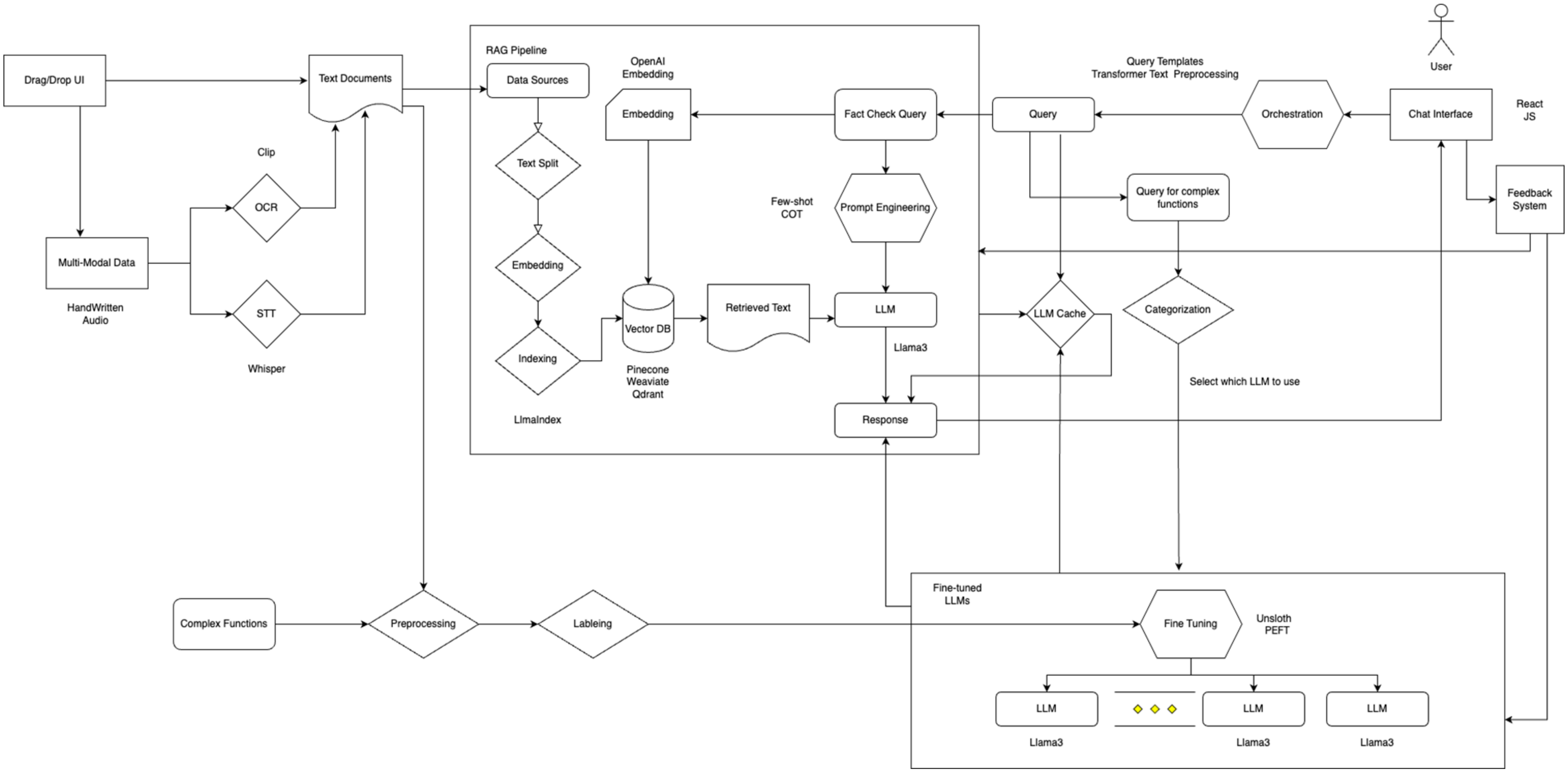
Multimodal data input

- Text, audio, and video content from government documents, reports, policies, and engagement sessions are transcribed, tagged, and indexed to reflect real-world and decision-making environments.

Continuous AI Alignment

- Learning cycles to update models with new state-specific knowledge, red-team testing, evaluation, and feedback, ensuring ongoing improvement, trustworthiness, and mission-aligned responses.

Solution Architecture



Development of process documentation for reproducibility across other states and for multiple use cases

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1
Set mission and value targets
Pick three high-value workflows per business line; define success measures and owners.

2
Select model approach
Local open source models for maximum control; managed private models for heavy or seasonal loads; combine both where sensible.

3
Establish policy and routing gateway
Enforce single sign on, role-based access, prompt redaction, spending limits, and full audit before any request reaches a model.

4
Collect and label knowledge
Ingest operating procedures, guides, contracts, catalogues, logs; label as public, internal, confidential, or restricted.

5
Build retrieval over your data
Create keyword and meaning-based search; enforce document permissions at retrieval time; store citations for every answer.

6
Secure the data path
Private networking only, customer-managed encryption keys, blocked outbound connections, and separation of duties.

7
Deploy the conversation interface
Self-hosted or custom branded chat inside your network with presets for each team and clear source citations.

8
Enable business tools
Safe read access to systems such as customer relationship management, ticketing, inventory, and plant historians; add confirmation for any write action.

9
Engineer for reliability
Use reproducible infrastructure, container orchestration, configuration in version control, and progressive rollout for models and retrieval rules.

10
Protect quality and safety
Require grounded answers with evidence, scan for toxic or personal content, and block disallowed tasks; run regular adversarial tests.

11
Control cost from day one
Right-size models, compress where possible, cache repeated work, batch similar requests, and set spending dashboards and alerts.

12
Pilot and measure
Launch to a focused group; track adoption, time saved, conversion lift, lower downtime, and cost per thousand tokens processed.

13
Integrate with systems of record
After proven wins, allow guarded write-backs such as draft orders, tickets, notes, and schedule updates with human approval.

14
Support local languages and offline work
Add local language materials and an offline mode that synchronises when a connection is available.

15
Govern retention and access
Enforce data labels in the interface, apply retention by role, run quarterly access reviews, and enable legal hold where required.

Product Demo

AIKano: Prompt Logic and Response Tuning Framework

Use Cases	Internal Prompt Engineering Logic	Response Tuning Requirements
Document Comparison & Evolution Tracking	Ask model to compare version A vs version B, extract all changes in wording, coverage, and scope; highlight additions, removals, shifts in intent.	Provide structured outputs: Change summary, thematic implications, timeline markers, confidence level. Avoid assumptions without retrieved citations.
Bias Detection & Inclusion Gap Analysis	Retrieve policies and scan for keywords, demographic mentions, absence of representation; major exclusions (e.g. displaced, female, rural, specially able etc.) apply sentiment analysis to language about identity groups.	Ensure neutral tone aligned to Kano ethno-religious guardrails; present findings as data-driven gaps, include mitigation suggestions, avoid normative statements or stereotypes.
Inter-agency Consistency Check	Compare policy language alignment across ministries; classify contradictions, overlaps, misaligned targets, or definitions.	Provide a 3-tier rating: aligned / partially aligned / conflicting; suggest points for harmonisation.
Regulatory Gap Identification	Retrieve sector policies and compare to current realities and standards; scan for outdated terminology, missing coverage, obsolete technologies.	Output gap list with why it matters, latest benchmark signals, and recommended content areas requiring update.
Data Extraction for Dashboards	Identify and extract quantitative fields and other named entities (location, issues, targets, budgets, KPIs) using structured parsing templates.	Return output as validated table format, enforcing units consistency; include placeholder for missing numerical fields.

Thank You

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